Predicting Mobile Review Sentiment from Specification and Review Features

A Supervised Learning Approach Using the Mobile Reviews: Sentiment and Specification Dataset (2025 Edition)

Submission 2: Proposal, Code, and Notebooks

Machine Learning Assignment — PS26 - DSC01 Machine Learning 120B

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Project Title	Predicting Mobile Review Sentiment from Specification and Review Features
Dataset Name	Mobile Reviews: Sentiment and Specification Dataset (2025 Edition)
Source	https://www.kaggle.com/datasets/mohankrishnathalla/mobile-reviews-sentiment-and-specification
Number of Rows	50,000+ (raw); ~47,500 (modeling subset)
Number of Columns	~25 (raw); 18 (engineered features)
Target Variable	sentiment_binary (binary): 1 if positive, else 0
Task Type	Binary Classification (Supervised Learning)
Number of Research Questions	7

1. Project Title

Predicting Mobile Review Sentiment from Specification and Review Features: A Supervised Learning Approach Using the Mobile Reviews Dataset (2025 Edition)

2. Dataset

Name: Mobile Reviews: Sentiment and Specification Dataset (2025 Edition)

Source: https://www.kaggle.com/datasets/mohankrishnathalla/mobile-reviews-sentiment-and-specification

Original size: 50,000+ rows x ~25 columns

Modeling subset: ~47,500 rows x 18 engineered features (after removing rows with missing sentiment, rating, or price)

Key columns used

- sentiment — the review sentiment label (Positive/Negative); used to define the binary target
- rating — numeric star rating (1-5); used as a feature and fallback target
- price — device price in USD; used to derive log_price and is_flagship
- review_text — raw review text; used to derive log_review_length (word count)
- brand — manufacturer name; used to derive 7 binary brand indicator features
- ram — RAM size in GB; used as a numeric feature
- storage — internal storage in GB; used as a numeric feature
- battery — battery capacity in mAh; used as a numeric feature
- screen_size — display diagonal in inches; used as a numeric feature
- camera_mp — primary camera resolution in megapixels; used as a numeric feature
- review_date — used to derive review_year and review_month

3. Target Variable

sentiment_binary (binary, 0 or 1)

Definition: sentiment_binary = 1 if sentiment == 'positive', else 0

Justification: Binary sentiment classification is the standard formulation in NLP and e-commerce review analysis. The positive/negative label directly captures user satisfaction and is actionable for both consumers and manufacturers. Using the dataset's pre-annotated sentiment column ensures consistency and reproducibility.

Class balance on the modeling subset: ~54% positive (n~25,650), ~46% negative (n~21,850). The dataset is near-balanced and does not require aggressive resampling.

4. Type of Task

Binary Classification (Supervised Learning).

5. Problem Statement

Mobile device reviews represent one of the most important consumer feedback channels in the global consumer electronics market. With over 1.4 billion smartphones sold annually, understanding what drives positive versus negative reviewer sentiment is valuable for manufacturers, retailers, and consumers alike.

This project addresses the question: given structured metadata about a mobile device (price, RAM, storage, camera, battery) and basic review properties (length, rating, brand, date), can we predict whether a review will express positive sentiment with high accuracy?

Unlike approaches that rely on raw review text (requiring NLP pipelines and large language models), this project focuses on structured tabular features that are readily available at scale and interpretable by non-technical stakeholders. The findings aim to identify which device specification and review characteristics most strongly predict positive sentiment, and whether predictability varies across price tiers, brands, and review lengths.

6. Research Questions

This project is guided by the following seven research questions, each addressing a distinct aspect of the supervised learning problem:

- RQ1: How consistent are model performance estimates between a single stratified hold-out split and 5-fold stratified cross-validation?
- RQ2: Do gain-based and permutation-based feature importance methods agree on which signals most drive sentiment prediction?
- RQ3: Does positive sentiment rate and model predictability vary systematically across device price tiers (Budget / Mid-range / Premium / Flagship)?
- RQ4: Does a model trained on the full dataset predict sentiment equally well across different mobile brands (Samsung, Apple, Xiaomi, OnePlus, Realme)?
- RQ5: How do RAM size and internal storage jointly affect positive sentiment rate and model predictability?
- RQ6: How do class imbalance mitigation strategies (SMOTE, undersampling, class weighting) affect the precision-recall trade-off?
- RQ7: Does review length tier (Short / Medium / Long / Very Long) influence sentiment predictability?

7. Proposed Methodology

Stage 1: Data acquisition and inspection

- Download the Mobile Reviews CSV from the public Kaggle source (50,000+ rows).
- Inspect schema, dtypes, and missingness patterns across all ~25 columns.
- Document data quality observations and identify columns suitable for feature engineering.

Stage 2: Filtering and target definition

- Remove rows with missing sentiment, rating, or price (~2,500 rows), leaving ~47,500 records.
- Define binary target: sentiment_binary = 1 if sentiment == 'positive', else 0.

- Verify class balance (~54% positive — near-balanced, minimal resampling needed).

Stage 3: Feature engineering

- Numeric transforms: log_price = log(1+price), log_review_length = log(1+word_count).
- Device specs: ram_gb, storage_gb, battery_mah, screen_size_inch, camera_mp (extracted numerically).
- Temporal features: review_year, review_month (from review_date).
- Price indicator: is_flagship = 1 if price > $700, else 0.
- Brand indicators: 7 binary dummies for Samsung, Apple, Xiaomi, OnePlus, Realme, Oppo, Vivo.
- Final feature matrix: 18 numeric features ready for sklearn-compatible models.

Stage 4: Train-test split

- Stratified 80/20 split using sklearn with random_state=42 for reproducibility.
- Stratification preserves the ~54% positive rate in both train and test sets.

Stage 5: Model training

- Five supervised classifiers: Logistic Regression, SVM (RBF), Random Forest, Gradient Boosting, XGBoost.
- Linear/kernel models receive standardised features (StandardScaler); tree-based models use raw features.

Stage 6: Evaluation

- Predictions evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Per-segment evaluation for RQ3 (price tiers), RQ4 (brands), RQ7 (review lengths).
- Four imbalance strategies compared on the same held-out test set for RQ6.

Stage 7: Visualisation and reporting

- Each RQ produces one publication-grade figure (PDF + PNG) and one results table (CSV).
- All seven notebooks are self-contained and runnable on Kaggle without modification.

8. Machine Learning Models

Model	Library	Family	Role

Logistic Regression	sklearn.linear_model	Linear	Interpretable linear baseline
SVM (RBF kernel)	sklearn.svm	Kernel	Non-linear baseline
Random Forest	sklearn.ensemble	Bagging	Variance-reducing tree ensemble
Gradient Boosting	sklearn.ensemble	Boosting	Sequential tree ensemble
XGBoost	xgboost	Boosting	State-of-the-art gradient boosting

9. Evaluation Metrics

Metric	Definition	Why it matters
Accuracy	(TP + TN) / total	Overall correctness; easy to interpret
Precision	TP / (TP + FP)	Of reviews predicted positive, how many truly are
Recall	TP / (TP + FN)	Of actual positive reviews, how many the model catches
F1-Score	Harmonic mean of precision and recall	Balances precision and recall; primary headline metric
ROC-AUC	Area under ROC curve	Threshold-independent ranking quality measure

10. Expected Figures and Tables

RQ	Figure	Table
RQ1	Grouped bar chart: hold-out vs 5-fold CV per model with error bars	table_rq1_cv_stability.csv
RQ2	Side-by-side horizontal bars: gain-based vs permutation importance	table_rq2_permutation_importance.csv
RQ3	Two panels: positive rate by price tier + model performance bars	table_rq3_price_tier_sentiment.csv
RQ4	Two panels: positive rate by brand +	table_rq4_brand_analysis.csv

	Accuracy/F1/AUC grouped bars	
RQ5	3x3 heatmaps: positive sentiment rate and F1 by RAM x Storage tier	table_rq5_ram_storage_interaction.csv
RQ6	Grouped bar chart: four imbalance strategies x five metrics	table_rq6_imbalance_strategies.csv
RQ7	Two panels: positive rate by review length tier + F1/AUC bars	table_rq7_review_length_tiers.csv

11. Tools and Infrastructure

- Programming language: Python 3.10+
- Numerical computing: NumPy, Pandas
- Machine learning: scikit-learn, XGBoost, imbalanced-learn
- Visualisation: Matplotlib
- Notebook environment: Jupyter (executable on Kaggle without modification)

12. Deliverables

- This proposal document (PDF + DOCX)
- Seven executed Jupyter notebooks (RQ1.ipynb through RQ7.ipynb) with visible outputs
- Seven figure PDFs and seven CSV tables produced by the notebooks
- README.md, Methodology.md, and requirements.txt
- Direct dataset link: https://www.kaggle.com/datasets/mohankrishnathalla/mobile-reviews-sentiment-and-specification